

Ideation Phase

Brainstorm & Idea Prioritization

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Team ID	6ab7f80ad083b0e2ee72b17e
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Team Members	Mehathaj Banu A; Haridharshini S; Yogalakshmi D; Hemavathy D; Malini SV
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization:

Brainstorming provides an open environment in which team members contribute different perspectives towards understanding student-learning challenges. The objective is to generate diverse solution directions, organize them into meaningful capability groups, and prioritize ideas using importance and feasibility.

For EduGenie, the activity focuses on how an AI-powered learning assistant can support students beyond basic question answering by adapting support to learner status, study context, learning format, progress and goals.

Step-1: Team Gathering, Collaboration and Select the Problem Statement

A. Team Gathering

The team has five members. Each member explored a distinct area so that the brainstorming activity covered different dimensions of the student-learning experience.

Team Member	Name	Brainstorming Focus Area
Member 1	Mehathaj Banu A	Adaptive Diagnostic Learning
Member 2	Haridharshini S	Smart Study Planning & Time Management
Member 3	Yogalakshmi D	Multimodal Learning Support
Member 4	Hemavathy D	Learning Progress Analytics & Goal Tracking
Member 5	Malini SV	Context-Aware Academic Assistance

B. Set the Goal

The brainstorming goal is to identify practical solution directions that can help students:

- understand their current learning status and knowledge gaps
- plan study activities around subjects, deadlines and available time
- learn through more than one representation when a single explanation is insufficient
- monitor progress and work towards measurable learning goals
- receive academic support that remains relevant to the current subject and study context

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C. Define Your Problem Statements

The team identified three core student problems. Each is expressed as a “How Might We...?” question so that it can directly guide solution ideation.

ID	Core Problem Statement	How Might We...?
PS-1	Limited Awareness of Current Learning Gaps	How might we help students identify what they already understand, what they have missed, and which topics require additional attention?
PS-2	Difficulty Organizing Study Time Around Academic Demands	How might we help students create realistic study plans that balance subjects, deadlines, priorities and available study time?
PS-3	Difficulty Tracking Learning Progress and Maintaining Context	How might we help students monitor their progress and receive relevant academic assistance that remains connected to their current subject and learning context?

Key Rules of Brainstorming

Stay focused	Encourage diverse ideas
Defer judgment during idea generation	Listen to others before evaluating
Build on useful suggestions	Keep ideas connected to student needs
Prefer clear and practical concepts	Record the reason behind important ideas

Problem Framing Note

The three problem statements are intentionally distinct: the first concerns **diagnosis**, the second concerns **planning**, and the third concerns **progress and contextual continuity**. Together they create a broader basis for designing EduGenie capabilities.

Problem-to-Ideation Focus

Core Problem	Primary Design Focus
PS-1	Discover the learner's present knowledge state and learning gaps.
PS-2	Convert academic workload and available time into an achievable study plan.
PS-3	Connect progress information with relevant, context-aware academic assistance.

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Brainstorm, Idea Listing and Grouping

Step-2: Brainstorm, Idea Listing and Grouping

A. Brainstorm

Each member proposed ideas related to the assigned focus area. The ideas below are deliberately different from the concepts in the reference document.

Member 1 Adaptive Diagnostic	Member 2 Study Planning	Member 3 Multimodal Support	Member 4 Progress Analytics	Member 5 Context-Aware Assistance
Identify prerequisite gaps before starting a topic.	Build a weekly study schedule from deadlines.	Present a topic using text plus a simple visual representation.	Track topic completion and pending work.	Detect the subject from the active learning context.
Use short diagnostic questions to estimate readiness.	Allocate time according to subject priority.	Convert dense notes into structured visual outlines.	Show learning milestones over time.	Keep follow-up answers connected to the current topic.
Flag topics where repeated mistakes occur.	Suggest realistic daily study blocks.	Offer alternative formats such as examples, diagrams and structured notes.	Record goals and progress checkpoints.	Use previous turns in the same session to avoid irrelevant responses.
Recommend a starting level from diagnostic evidence.	Rebalance the plan when a task is delayed.	Let students choose the representation that helps them most.	Highlight progress toward a learner-defined target.	Tailor examples to the academic subject being studied.
Generate a learning-readiness summary.	Prioritize urgent academic tasks.	Combine multiple representations for complex material.	Provide a simple progress dashboard.	Give subject-specific guidance while preserving context.

B. Group Ideas

Group 1: Adaptive Learning Diagnosis

Purpose: Identify a student's present knowledge state and use evidence of understanding to determine suitable starting points and learning gaps.

- Diagnostic readiness checks.
- Knowledge-gap identification.
- Prerequisite detection.
- Error-pattern awareness.
- Learner readiness summary.

Group 2: Smart Study Planning

Purpose: Help students organize academic work into realistic study activities using priorities, deadlines and available time.

- Deadline-aware planning.
- Subject prioritization.
- Daily study blocks.
- Plan adjustment after delays.
- Workload balancing.

Group 3: Multimodal Learning Representation

Purpose: Present academic information through complementary representations so that students can select an appropriate way to process a topic.

- Structured visual outlines.
- Text-to-diagram transformation.
- Worked examples.
- Alternative explanation formats.
- Representation selection.

Group 4: Progress Analytics & Goal Tracking

Purpose: Make learning progress visible by connecting completed work, milestones, pending topics and learner-defined goals.

- Topic completion tracking.
- Milestone monitoring.
- Goal checkpoints.
- Pending-work visibility.
- Progress summaries.

Group 5: Context-Aware Academic Assistance

Purpose: Keep EduGenie assistance relevant to the student's active subject, topic and ongoing learning context.

- Subject-context detection.
- Session continuity.
- Context-relevant examples.

- Topic-linked follow-up support.
- Subject-specific guidance.

Ideation Phase

Idea Group → Problem → Capability Mapping

C. Idea Group → Problem → Capability Mapping

Each idea group supports one of the three core problems. The mapping shows how the brainstormed directions can become concrete EduGenie capabilities.

Idea Group	Core Problem	EduGenie Capability
Adaptive Learning Diagnosis	Problem 1	Diagnostic Learning Readiness & Gap Detection
Smart Study Planning	Problem 2	AI-Assisted Study Plan Generation
Multimodal Learning Representation	Problem 1	Multi-Format Learning Presentation
Progress Analytics & Goal Tracking	Problem 3	Learning Progress & Goal Monitoring
Context-Aware Academic Assistance	Problem 3	Context-Aware Academic Support

Overall Student Need → Capability Relationship

Student Problem	Student Need	Idea Groups	EduGenie Capability
PS-1: Limited Awareness of Current Learning Gaps	Know the current level of understanding and identify gaps.	Adaptive Learning Diagnosis + Multimodal Learning Representation	Diagnostic Learning + Multi-Format Support
PS-2: Difficulty Organizing Study Time	Know what to study, when to study it, and how to adjust the plan.	Smart Study Planning	AI-Assisted Study Planning
PS-3: Difficulty Tracking Progress and Maintaining Context	Monitor progress and receive assistance connected to the current academic context.	Progress Analytics & Goal Tracking + Context-Aware Academic Assistance	Progress Monitoring + Context-Aware Support

Capability Design Interpretation

- **Diagnosis:** EduGenie first estimates where the learner currently stands rather than assuming the same starting point for every student.
- **Planning:** Academic tasks can be converted into an organized schedule that reflects actual time constraints.
- **Representation:** Learning content can be presented in complementary forms rather than relying on one format.
- **Progress:** Goals and milestones provide a visible connection between activity and learning progress.
- **Context:** Assistance remains tied to the subject and topic being studied, improving relevance within a learning session.

This mapping establishes a traceable path from **student problem** → **student need** → **idea group** → **EduGenie capability**, which can be used during subsequent design and implementation stages.

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Idea Prioritization

Step-3: Idea Prioritization

The team placed the five idea groups on an Importance vs Feasibility grid. Importance represents the potential contribution to the identified student needs, while feasibility represents practical implementation considerations such as effort, time, integration and data requirements. No numerical scoring is used.

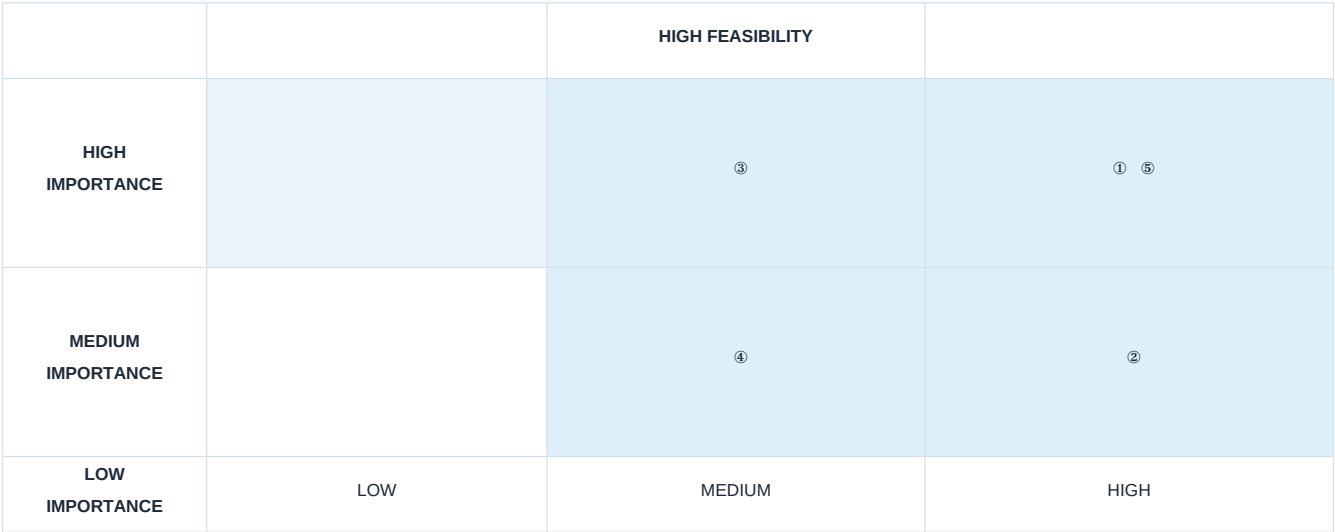


Figure 1: Importance vs Feasibility matrix for the EduGenie idea groups

No.	Idea Group
1	Adaptive Learning Diagnosis
2	Smart Study Planning
3	Multimodal Learning Representation
4	Progress Analytics & Goal Tracking
5	Context-Aware Academic Assistance

Prioritization Summary

Idea Group	Importance	Feasibility	Prioritization Rationale
Adaptive Learning Diagnosis	High	High	Directly addresses the need to identify learning gaps and can be supported through diagnostic interactions and learner-response evidence.
Smart Study Planning	High	High	Addresses a common planning need and can be implemented through deadline, workload and availability inputs.
Multimodal Learning Representation	High	Medium-High	Provides additional learning flexibility, while requiring careful design of consistent text, visual and structured representations.

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Idea Prioritization & Final Brainstorming Outcome

Idea Group	Importance	Feasibility	Prioritization Rationale
Progress Analytics & Goal Tracking	High	High	Makes learning activity and goals visible, while basic milestone and completion tracking can be implemented with manageable system logic.
Context-Aware Academic Assistance	High	High	Improves the relevance of academic support by connecting responses with subject, topic and session context.

Prioritization Interpretation

- **Adaptive Learning Diagnosis:** Prioritized because it establishes evidence about the learner before subsequent assistance is provided.
- **Smart Study Planning:** Prioritized because planning directly connects academic demands with the student's available study time.
- **Multimodal Learning Representation:** Kept as a strong direction because different representations can complement one another, although broader multimodal design requires additional implementation effort.
- **Progress Analytics & Goal Tracking:** Prioritized because visible progress supports continuous monitoring of academic work.
- **Context-Aware Academic Assistance:** Prioritized because contextual relevance is important when students ask follow-up questions during a learning session.

Final Brainstorming Outcome

The brainstorming activity resulted in five solution directions that support the three core student problems. These directions are not separate problem statements; they represent complementary capabilities that can be combined within the EduGenie learning assistant.

No.	Solution Direction	Core Problem	EduGenie Capability
1	Adaptive Learning Diagnosis	Problem 1	Diagnostic Learning Readiness & Gap Detection
2	Smart Study Planning	Problem 2	AI-Assisted Study Plan Generation
3	Multimodal Learning Representation	Problem 1	Multi-Format Learning Presentation
4	Progress Analytics & Goal Tracking	Problem 3	Learning Progress & Goal Monitoring
5	Context-Aware Academic Assistance	Problem 3	Context-Aware Academic Support

Outcome Statement

The prioritized directions provide a balanced ideation base: EduGenie can first understand the learner, help organize learning activity, present information flexibly, track progress, and maintain relevant academic context.

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Final Ideation Summary

Final Brainstorming & Prioritization Conclusion

The ideation phase established a structured set of solution directions for EduGenie: an AI-powered learning assistant designed around the changing needs of individual students.

The brainstorming process moved from student problems to focused solution directions. Instead of treating every idea as an independent feature, the team grouped related concepts and mapped them to the three identified core problems. This creates a clear connection between the learner's need and the capability proposed for the system.

Selected Solution Directions

Solution Direction	Primary Contribution
Adaptive Learning Diagnosis	Identifies learner readiness, prerequisites and knowledge gaps.
Smart Study Planning	Organizes academic tasks around priorities, deadlines and available time.
Multimodal Learning Representation	Provides complementary ways to represent and process learning content.
Progress Analytics & Goal Tracking	Connects learning activity with milestones, goals and pending work.
Context-Aware Academic Assistance	Keeps academic support relevant to the student's current subject and session context.

Design Traceability

The final ideation set maintains the following traceable sequence:

Stage	Result
Student Observation	Identify a concrete learning difficulty.
Problem Framing	Convert the difficulty into a focused "How Might We...?" question.
Brainstorming	Generate multiple possible approaches from different team perspectives.
Grouping	Combine related ideas into coherent capability areas.
Prioritization	Compare importance and feasibility without numerical scoring.
Outcome	Carry the selected solution directions into the next project phase.

Closing Statement

The Ideation Phase provides the project team with a clear and defensible foundation for the next stage of EduGenie development. The resulting solution directions are connected to identified student needs, organized into capabilities, and examined for practical implementation considerations.

Prepared by: Mehathaj Banu A, Haridharshini S, Yogalakshmi D, Hemavathy D, Malini SV